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This story is the second part of another story:

Information: Study from Sequence Analysis

The meta-alphabet

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Many people struggle to understand entropy because its foundation is not clear. Where does it come from? Here, entropy is defined as follows:

- It should be **quantitatively** proportional to the **most common state** (higher entropy means higher probability of positive cases).
- **Additivity:** If we join two systems, the entropy of the combined system should equal the sum of the individual entropies.
- It should be an analytical function, so that it can be used in mathematical operations; otherwise, it would have no practical use.

Shannon's entropy formula meets these requirements, but it is essentially an approximation valid for large numbers of states. For macroscopic particle systems with billions of states, the approximation works well. However, it fails for systems with only a few bits.

The new definition

If the goal is simply to identify the most common state, we can use the coefficient of the respective combination. For example, with 3 bits, the states with coefficients of 3 are the most common (high entropy). For 8 bits, the highest entropy comes from states with coefficients of 70.

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$$x^8 + 8x^7 + 28x^6 + 56x^5 + 70x^4 + 56x^3 + 28x^2 + 8x + 1$$

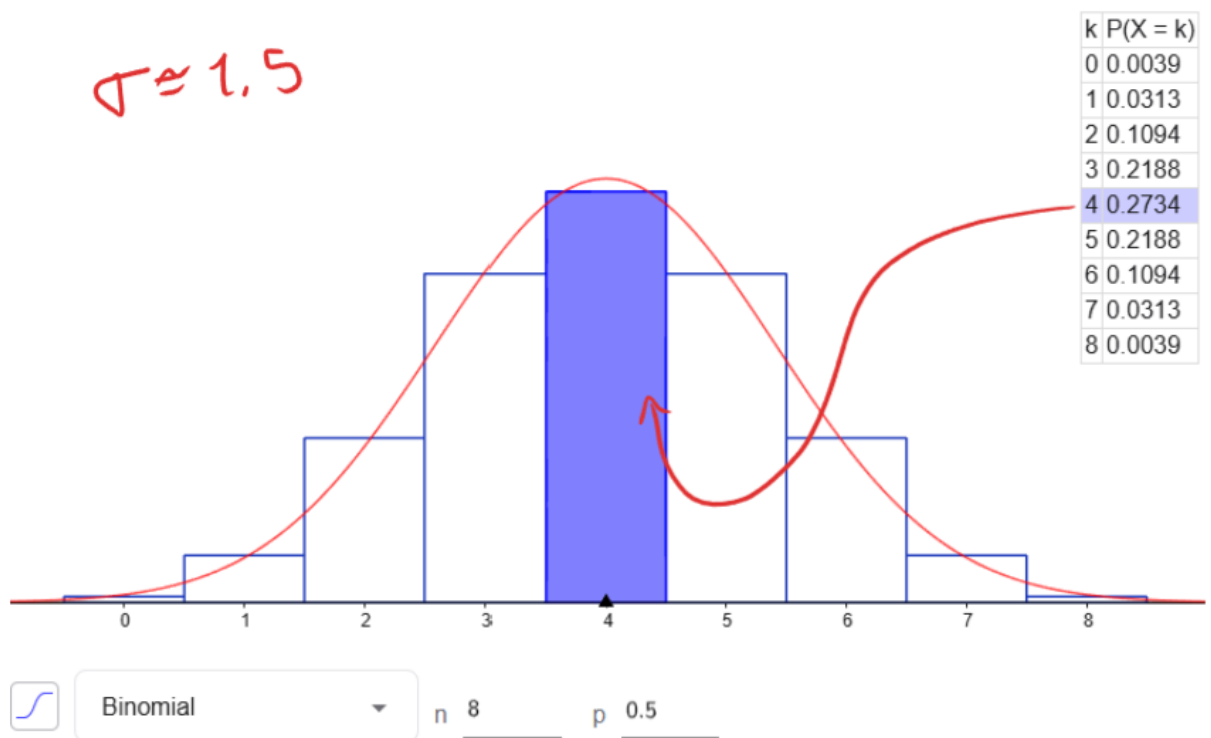
Higher
entropy
state

Why? Because the most common state is the one most likely to occur when chosen randomly.

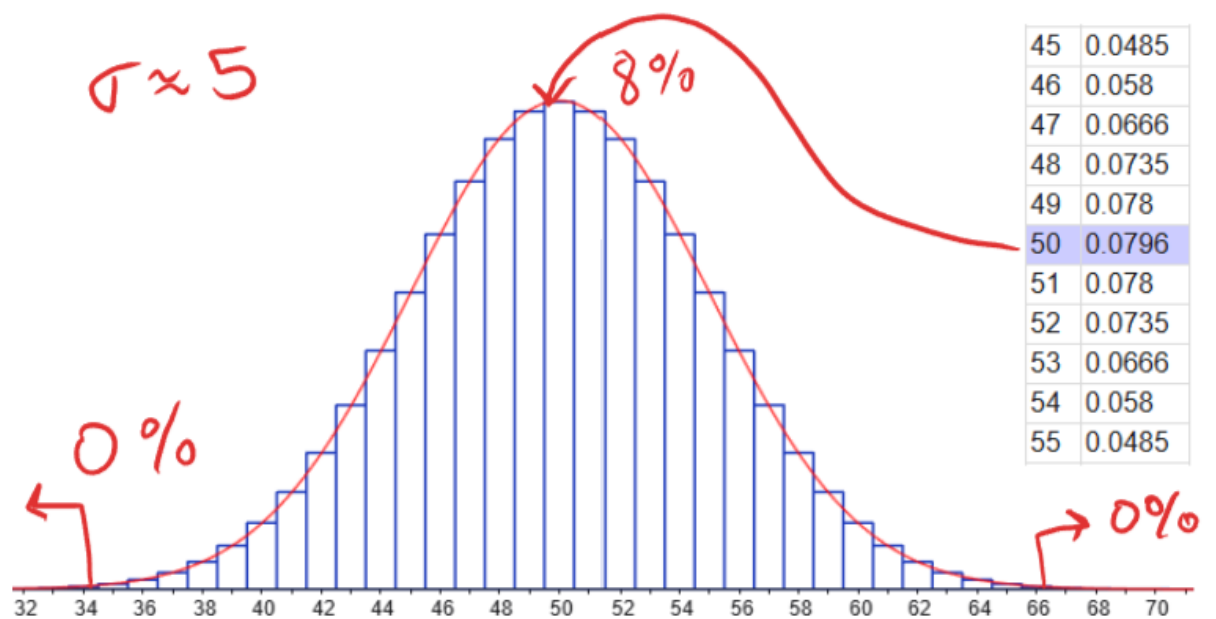
For 8 bits, the probability of the state of 4 ones and 4 zeros is about 27.34%. With larger systems, such as 100 bits, the probability decreases to around 8%, but the principle remains: entropy points to the most expected state, which lies at the average of the distribution.

Okay, no, it is decreasing, the reason is the normalization, yet the idea here is that the most expected state is given by the distribution (See graphs, similar to a Gaussian distribution in red line). I guess we should do a confidence zone (2–3 standard deviations).

Let's see in the graph what happens to $(1+x)^8/2^8$
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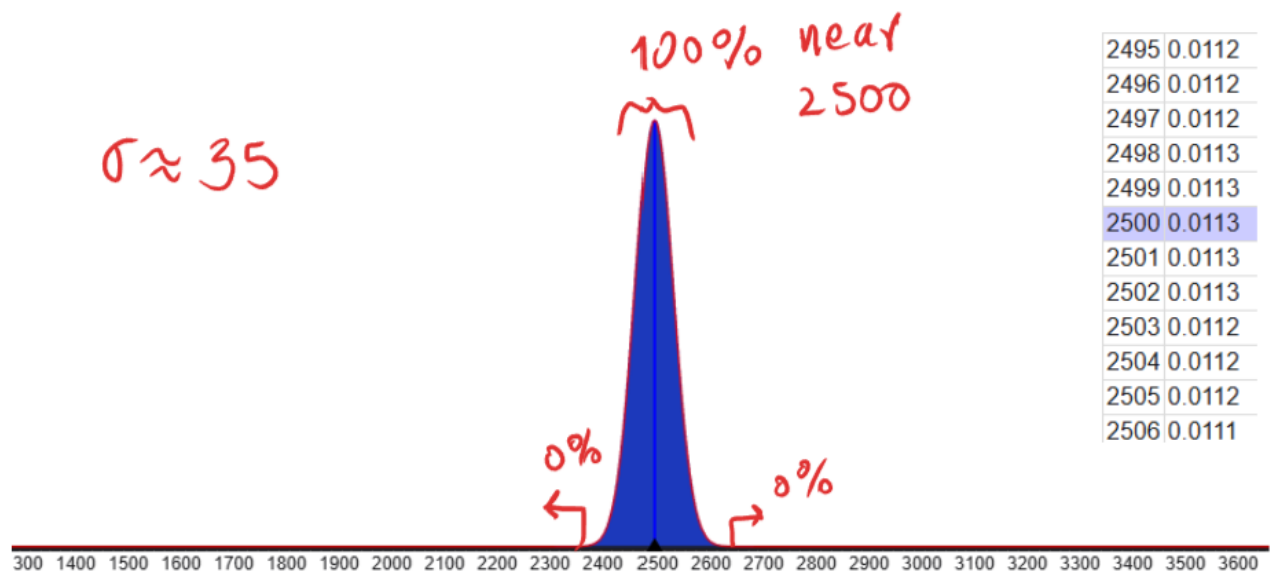


Now with $(1+x)^{100}/2^{100}$:
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Last example with 5000:

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Even after normalization, 2500 accounts for only 1.13%, but the distribution narrows around 2500, so it effectively becomes 100% near that number, which shows that using it is a good approximation for large numbers.

The point here is that entropy's purpose is mainly to tell us what state is most common; that is all. We can calculate the coefficient by the expansion of the binomial, or directly by using $nCr(N, N/2)/2^N$, in the example of 100 bits, it is $nCr(100, 50)/2^{100}$.

The new definition of entropy is that nature itself is an expression of randomness, and we can digitalize randomness in yes/no questions, then we can describe nature with the binomial expansion. This means the “true” entropy is:

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$$\text{Entropy} = \binom{D}{D/2}$$

$D := \text{number of bits}$

New definition of entropy

In practice, entropy is mainly used in optimization problems: maximizing entropy over a system means identifying the state that is most common among all possible configurations. This allows us to ignore the normalization term, as with it or without it, the maximum gives the same result.

Alternatively, we can also apply log to it, and it keeps the same maximum:
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$$\text{Entropy} = \ln\left(\binom{D}{D/2}\right)$$

The goods and the bads of this definition

Advantages: This definition makes entropy more intuitive. It highlights not only the most common state but also the distribution around it. It also works at the quantum level, where Shannon's entropy would fail because the approximation is not good for small numbers of bits, instead using the distribution would be desired.

Limitations: Although analytical, the combinatorial approach is difficult to work with. For example, adding entropies or differentiating the formula is hard. Approximations (such as using the Gamma function or Stirling's approximation) help bridge the gap, ultimately showing that Shannon's entropy is a special case of this generalization. There is actually some proof around internet of this.

The exploration

What comes next is a try at **reverse engineering**. I didn't know this connection, and I was exploring to improve this new definition of entropy instead of trying to be connected with the shannon's entropy.

Fixing problem 1: Add two entropies

Remember that we are interested in a definition of entropy that also has the additivity property, so let's check:

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$$\underbrace{3 \text{ bits}} + \underbrace{5 \text{ bits}} = \underbrace{8 \text{ bits}}$$

Entropy $\rightarrow \underbrace{\binom{3}{1.5}}_{\approx 3} + \underbrace{\binom{5}{2.5}}_{\approx 10} \stackrel{?}{\neq} \underbrace{\binom{8}{4}}_{\approx 70}$

in probability:

$$\frac{3}{2^3} + \frac{10}{2^5} = \frac{3 \cdot 2^2 + 10}{2^5}$$

$$= 176 \quad \text{vs} \quad 70$$

The reason why using combinatorics to define entropy doesn't add up well is that it is the number of possible cases in each system individually; it doesn't count the mixing combination between the systems. It should be:

the_number_of_cases_systemA + the_number_of_cases_systemB + the_number_of_cases_systemA_AND_B.

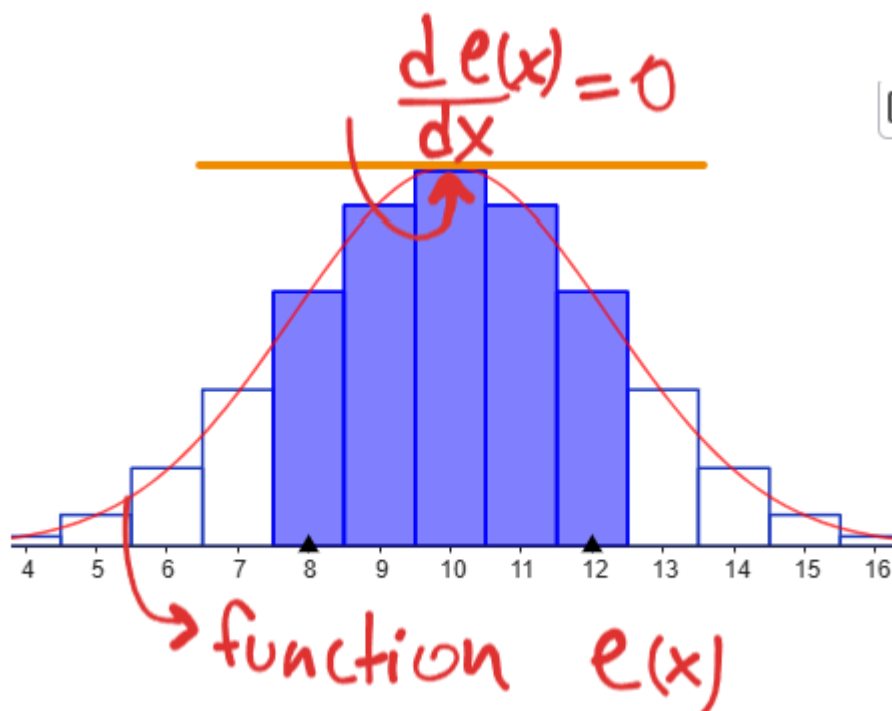
Theoretically, it can be calculated as simply adding the total slots and choosing half...

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Entropy $\rightarrow$

Fixing problem 2: Derivative

Without constraints, we already know the most common state is when x is maximum here: $nCr(100,x)$. We can get the most common states by finding the derivative equal to zero, due to this, we are interested in entropy definitions with easy derivatives.

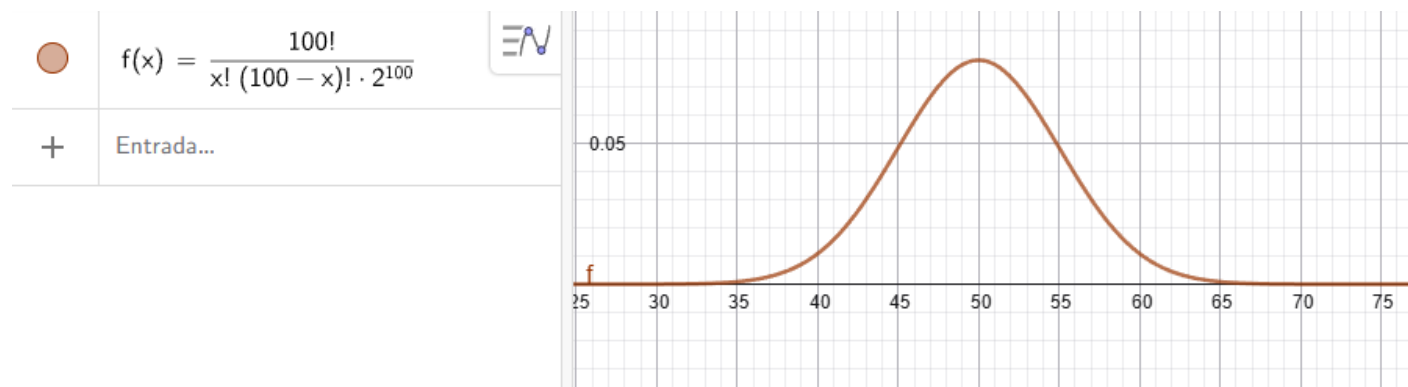


But what is the function $e(x)$? can be easily differentiated? Well, to begin with, the combinatory formula is discrete, and analytic functions need to be continuous. The good news here is that the combinatory formula can be generalized to real numbers using the **Gamma function**. For the purposes of this story, we can assume $nCr(100,x)$ where x is a real number.

$$e(x) = \frac{n!}{(n-x)! \cdot x!} = \binom{n}{x}$$

Plotted for $n=100$ is:

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*Failed tries (optionally skip it until “*End of failed tries“):

Now we want to derive this, which is hard. A first approach is to derive x factorial, which is also hard, but we can use a cached familiar function, the digamma function:

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$$\frac{d(x!)}{dx} = x! \underbrace{\psi(x+1)}_{\frac{d(x!)}{x!}}$$

digamma
function

$$\psi(x+1) = \frac{d(x!)}{x!}$$

Now with the other term $(n-x)!$, due to the chain rule is quite similar:
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$$\frac{d((n-x)!)}{dx} = -(n-x)! \psi(n-x+1)$$

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$$\frac{d}{dx} \frac{n!}{f \circ g} = n! \left(\underbrace{\frac{-f'}{f^2}}_{-\frac{\Psi(x+1)}{x!}} + \underbrace{\frac{-g'}{g^2}}_{-\frac{\Psi(n-x+1)}{(n-x)!}} \right)$$

If we equalize it to zero and try to find which 'x' gives the answer, it is still hard stuff, but it may exist in an easier way (Stirling approximation).

Not quite adding yet, though some are similar to Shannon's entropy. I came up with this other approximation by using the Taylor expansion:

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$$\frac{d(x!)}{dx} \approx x! \underbrace{\Psi(x+1)}_{\ln(x) + \frac{1}{2x} + \dots} \approx x! \ln(x)$$

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$$\frac{d}{dx} \frac{n!}{x!(n-x)!} = -n! \left(\frac{\ln(x)}{x!} + \frac{\ln(n-x)}{(n-x)!} \right)$$

Remember that the Shannon formula is:
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$$\text{Entropy} = - \sum_x \underbrace{p(x)}_{\text{Probability of symbols}} \log_2 \left(\underbrace{p(x)}_{\text{Probability of symbols}} \right)$$

Shannon's entropy

Until now, look at the similarity.
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$$\begin{aligned} \frac{d}{dx} \frac{n!}{x!(n-x)!} &= -n! \left(\frac{\ln(x)}{x!} + \frac{\ln(n-x)}{(n-x)!} \right) \\ \Rightarrow \sum \frac{\ln p_i}{p_i!} &= \frac{\ln(p_1)}{p_1!} + \frac{\ln(p_2)}{p_2!} \end{aligned}$$

By looking at its form of $\ln(x)$, we can use it from the beginning:

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$$\ln(e(x))$$

$$\ln\left(\frac{n!}{(n-x)!x!}\right) = \ln(n!) - \ln((n-x)!) - \ln(x!)$$

I didn't realize that the $\ln()$ was by definition and not by consequence of the derivative of the combinatorial formula. The new definition of entropy is obtained by applying $\ln()$, which keeps the same maximum, and also improves additivity

Now, if we derive the new definition of entropy, then $\ln(n!)$ doesn't matter; it becomes 0:

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$$\ln(e(x))$$

$$\frac{d}{dx} \ln\left(\frac{n!}{(n-x)!x!}\right) = \cancel{\ln(n!)} - \underbrace{\cancel{\ln((n-x)!)}}_0 - \underbrace{\cancel{\ln(x!)}}_0$$

$$\frac{1}{(n-x)!} \cdot \psi(n-x+1) - \frac{1}{x!} \cdot \psi(x+1)$$

The thing that is missing (and that makes the approximation valid, not just an approximation) is the tendency to a big number. Remember, the Shannon entropy is for a large number of states.

*End of failed tries

Let's forget the derivative for now, and only focus on having the same maximum.

I will cheat and use an approximation I made up for this:

$$\ln(x!) = x \ln(x)$$

With this approximation, you will see the relation between the combinatory formula and Shannon's entropy: Before a redefinition to match $(N-x)$ as a probability.

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$$\lim_{N \rightarrow \infty} \frac{N!}{\underbrace{(N-x)!}_{q!} \underbrace{x!}_{p!}}, \quad \underbrace{x \rightarrow xN}_{\text{normalize } x \in [0, 1]}$$

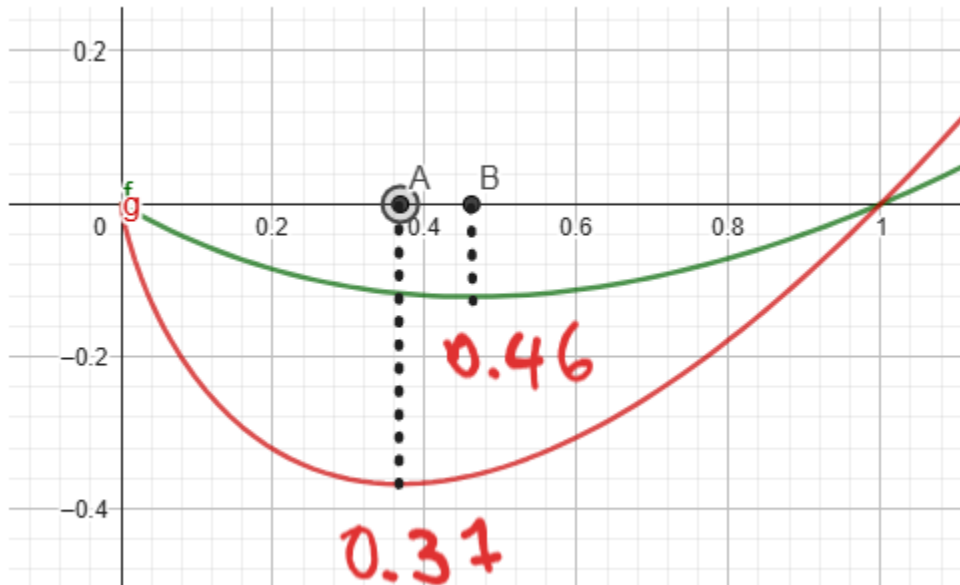
$$\Rightarrow \frac{N!}{((1-x)N)! (xN)!} \xrightarrow{\text{same max}} \frac{1}{(1-x)! x!}$$

And if we apply the derivative to the above expression, we just got
 Shannon's formula for entropy from the combinatorial formula:
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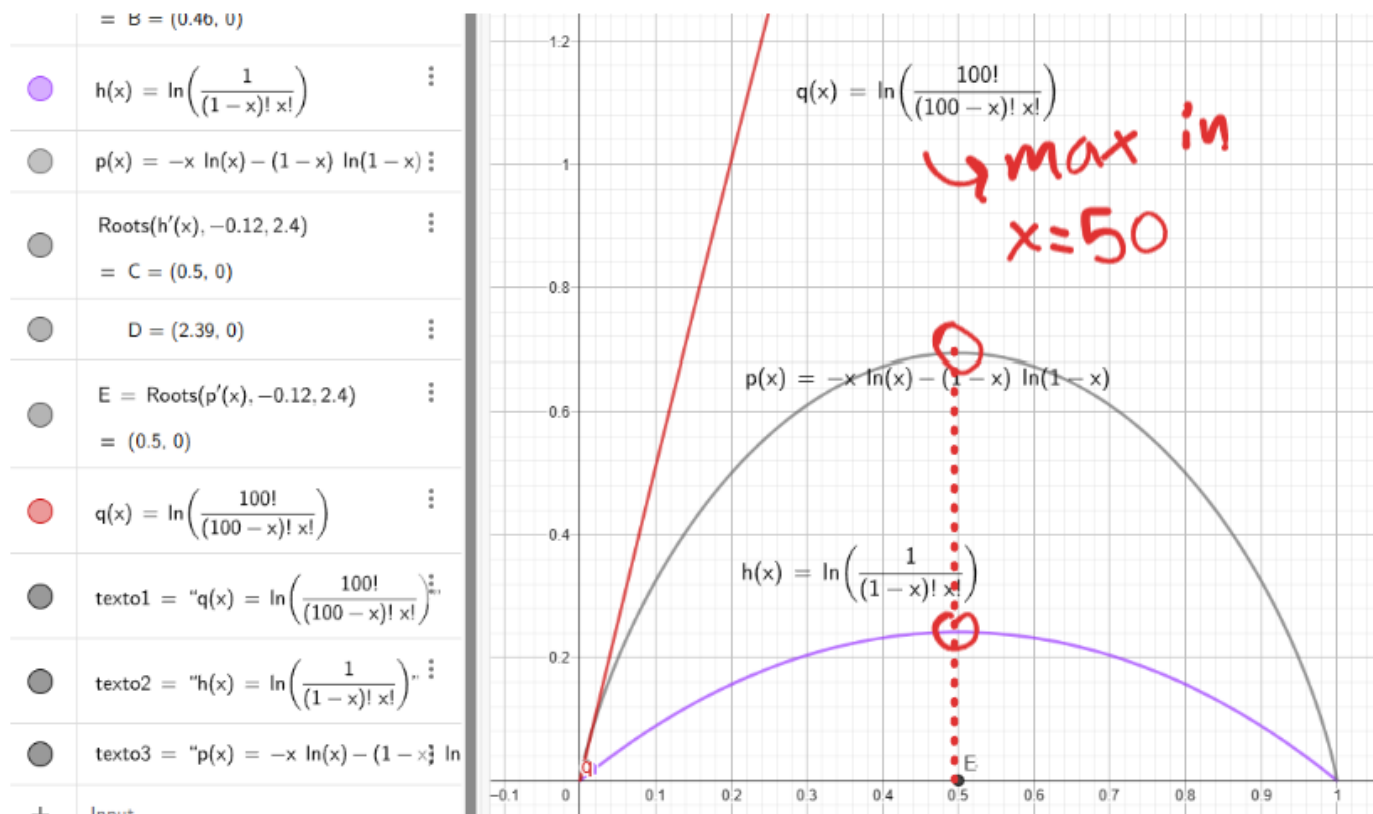
$$\ln \left(\frac{1}{(1-x)! x!} \right) = -\ln((1-x)!) - \ln(x!)$$

$$\text{Cheat} \Rightarrow \underbrace{-(1-x)\ln(1-x) - x\ln(x)}_{-\sum p_i \ln(p_i)}$$

How good is this approximation? When we plot both and try to find the maximum we get this difference:



Remember, what we care about is just the point where it maximizes; there is an error of 19%, yet both tend to select a value in the middle. **I did wrong**, above approximation is not good for $x \cdot \ln(x)$, but it preserve the same maximum for $\ln(nCr(N, N/2))$
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So the error is 0% for the maximum location... I forgot the $(1-x)\ln(1-x)$ part.

Recapitulating: We forget about the derivative and only focus on keeping the same maximum.

It's done, now there is no problem, but the new definition of entropy (I will call it a generalization of Shannon's entropy, because Shannon's entropy meets the same maximum when N tends to infinity).

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Shannon's entropy Generalization

$$\begin{aligned}\text{Entropy} &= \ln\left(\frac{1}{p_1! p_2! p_3! \dots}\right) \\ &= -\sum_i \ln(p_i!)\end{aligned}$$

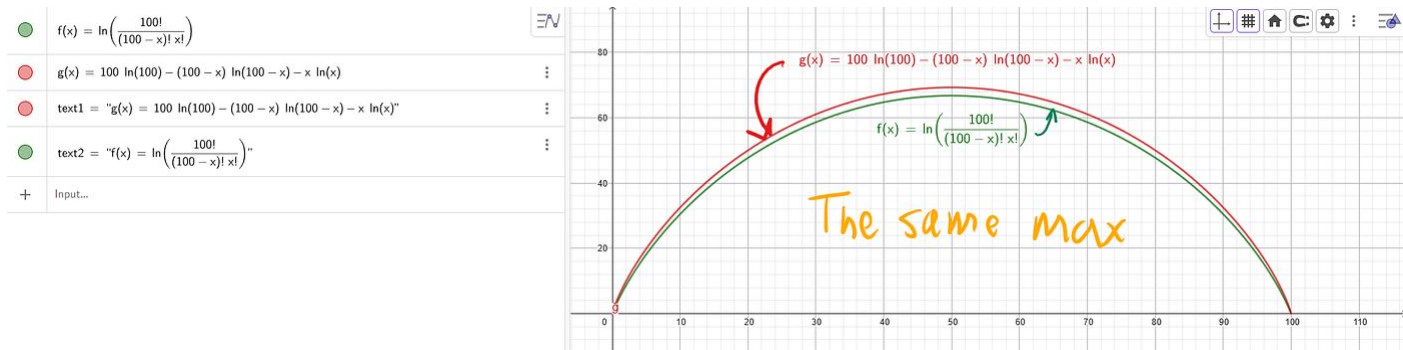
Shannon's case

if #slots $D \rightarrow \infty$

$$\Rightarrow -\sum_d p_d \ln(p_d)$$

After this, I was wrong when I thought Shannon's entropy formula didn't work for a few-bit systems; the maximum gives the same location no matter the number of bits. Plot check:

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The good news about this generalization is that Shannon's formula for entropy was intuitively derived from observation, while this re-definition has a theoretical basis. Making it possible to do a better interpretation of what entropy is.

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$$\text{Entropy} = \ln(p_1! \cdot p_2! \cdot p_3! \cdots)$$

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$$\text{Entropy} = -\sum \ln(p_i!)$$

More advantages are that with $\ln(x)$, entropy becomes simply additive:

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$$\underbrace{\text{Entropy}_1 + \text{Entropy}_2}_{\sum \ln(p_i) + \sum \ln(q_i)}$$

bits of info
bits of info

Why this approximation? $\ln(x!) \sim x \ln(x)$

The purpose of this approximation is not to make both functions $x!$ and $x \ln(x)$ tend to be equal. The goal is to make them the same maximum.

YET, this approximation is not simply taken from a hat; it has a reason. BUT keep in mind, even without a reason, it is true that both have the same maximum location, and that is all we care about in terms of defining entropy.

Let's go to understand why I thought $\ln(x!) \sim x \ln(x)$ was a good idea:
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$$x! = x(x-1)(x-2)\dots$$

if $x \gg 1 \Rightarrow x \approx x-1 \approx x-2$
large numbers $9999 \approx 9998$

$$\Rightarrow \lim_{x \rightarrow \infty} (x!) \approx \lim_{x \rightarrow \infty} x^x$$

Then, if you just apply $\ln()$ to both sides:

$$\ln(x!) = x \ln(x)$$

$\rightarrow$ only if $x \sim \infty$ log property

This may not be valid for a few bits, yet this approximation is valid for x greater than $\alpha \cdot \omega$, but for any 'x', it meets that the maximum location of the two definitions of entropies is the same location. So, the approximation $\ln(x!) = x \ln(x)$ is good for x transfinite, but even if x is not transfinite, this approximation results in the same answer to the maximum problem.

And as you saw far above, the original combinatory for a lot of bits has $(x \cdot N)!$ not $x!$, from $(x \cdot N)!$ You can appreciate $x \cdot N$ is a transfinite number if

N is. So the $\ln((x*N)!)=(x*N)*\ln(x*N)$ is totally valid. also $(N-x)!$ because it is $((1-x)N)!$ another transfinite number.

The use of transfinite numbers here is only to ensure $x*N$ is really big.

*Also found out later, this is quite similar to Stirling's approximation:
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This is [Stirling's series](#) with only the first term retained and, for large n , it reduces to Stirling's approximation

$$\ln n! \approx n \ln n - n.$$

(25)

Source: [Stirling's Approximation — from Wolfram MathWorld](#)

Recapitulation:

We used transfinite numbers at $\ln((x*N)!)=(x*N)*\ln(x*N)$ if we have a lot of bits (in this case, N tending to infinity, previously was $d \sim$ digits). By using it, we get that for transfinite numbers (let's say really big numbers) the $\ln(p_n!)$ matches Shannon's entropy formula, but also that we are building this entropy formula from the beginning from the binomial theorem.

We found that given the new definition of entropy, Shannon's entropy is a special case. Hence, I call this re-definition "The generalization of Shannon's entropy".

In particular, Shannon's entropy is equivalent to the new entropy formula, but for transfinite numbers (really big numbers). Whether or not they are transfinite numbers or not, both entropy formulas yield the same maximum location.

Warning! There is something a reader shared in a previous story, to be careful with $1 = (1)*(1-1/N)*(1-2/N)*(1-3/N)*\dots$ I states it tends to 1, but for $n \rightarrow N$ we can check that $(1)*(1-1/N)*(1-2/N)*(1-3/N)*\dots*(1-N/N)$, which is multiplying by zero then $N/N \dots$ The issue here is approximations:
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$$\lim_{N \rightarrow \infty} \prod_{n=0}^m \left(1 - \frac{n}{N}\right) = \begin{cases} 1, & \text{Real } m < \alpha N \\ 1 - \alpha, & m = \alpha N \text{ transfinite} \end{cases}$$

This is a cheat, yet it is something I am currently learning about. It seems this limit is 1 for m tending to $m/N=0$, and different than 1 when m is big enough such that $\alpha=m/N$ (α different than 0). So ' m ' tends to an infinity, but that infinity is less than $\alpha \cdot N$. My current interpretation is that it is 1 as long as ' m ' is real and not transfinite. I guess the last number before transfinite $N \sim N-1, \dots$ I may find out the answer another day.

btw spoiler about this, using Euler reflection formula $1/((-x)!(x!)) = \sin(\pi x)/(\pi x)$
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$$\frac{1}{(1-x)! \cdot x!} = \frac{\sin(\pi x)}{\pi x (1-x)}$$

transfinite answer

$x \sim \frac{\alpha}{N} \rightarrow$ real number answer

$\Rightarrow \frac{1}{1! \cdot 0!} = 1 = \lim_{n \leq \alpha N} \pi \left(1 - \frac{n}{N}\right)$

Next related story:

Another thing I wanted to share is to send messages not through configurations, but by **generating functions**, which means using frequencies. Example: 10% of ones means a message, 20% means another message, and so on... This way we can encode secret messages with stochastic behaviour, but also, we can send messages without the need to know all bits in the sequence, and just a chunk of bits will be enough.

Sending messages with something that looks random. This could explain why nature forms patterns and fractal patterns, also the “Law of Increasing Functional Information,” which is the generalization of the “evolution theory”.

Thanks for reading!

Wait, lastly, what is the interpretation of this formula?
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$$\ln((xN)!) = (xN) \ln(xN)$$

$$\rightarrow (xN)! \approx (xN)^{xN} \rightarrow \text{\#bits}$$

Permutations
\# Symbols

$$\Rightarrow \text{\#bits} \cdot \ln(\text{\#symbols})$$

The connection between the combinatory formula and Shannon's entropy is given by the above expression given transfinite numbers. And it comes from $\ln(nCr(N, N/2))$, which, in an abstract sense, is the logarithm of the configurations of the most common state. If we write the number of configurations given a basis, for example, base-2, we can write 128 as 2^7 , then its logarithm becomes $7 \cdot \ln(2)$. This takes the form of $\text{slots} \cdot \ln(\text{symbols})$

If we use log base-2, then $\log(2^7) = 7$, which tells us directly the amount of bits of information. In this sense, $\log(x!)$ counts the bits necessary to do $x!$ messages. And for transfinite numbers, $\ln(x!) \sim x \cdot \ln(x)$, so the entropy is counting the "bits" (nats if log base-e) of the number of configurations.

Whether it is information or chaos... as the original formula is $\ln(N!/((N-x)!x!))$, we can appreciate from here it becomes $\ln(N!) - \ln((N-x)!) - \ln(x!)$ equivalent to $\text{bits_information} - \text{bits_prob1} - \text{bits_prob2}$
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Entropy

$$\ln\left(\frac{N!}{(N-x)!x!}\right) = \underbrace{\ln(N!)}_{\text{total space (bits)}} - \underbrace{\ln((N-x)!)}_{\text{pattern1 information}} - \underbrace{\ln(x!)}_{\text{pattern2 information}}$$

Rate of information by the patterns

Example, if $x=N$, then total info= $\ln(N!)$, info of pattern = $-\ln(1)-\ln(N!)$... We had $\ln(N!)$ bits to use, and the pattern use $\ln(N!)$ of them, $\ln(1)=0$, as $nCr(N,N)=1$

Conclusion

Even if the process is wrong, including the cheat formula and the wrong use of transfinities, the new entropy definition provided meets the 3 requirements initially postulated.

1. It quantifies the most common microstate, then at maximum it does the same work as the shannon's entropy
2. It is easily additive; we can add the entropies of different systems just by addition. The interpretation is adding bits of information.
3. **(fails)** It is not really easy to handle, yet as we know, the maximum is the same, and that this derivation of entropy comes from the binomial theorem and randomness, then I would prefer this as the "original" entropy, because the shannon's entropy, as usually shown, is heuristic (not a solid background other than trying it and working). Yet, although shannon's entropy gives the same maximum, using the binomial expansion is more descriptive as it shows the entire distribution and not only the most common configuration.

Many tries around finding the relation between both, not sure if successfully done, the use of transfinite here needs more study. Might zeros in $(1-n/N)$ cancel out with some infinities. This is usually not allowed, but when working with limits and transfinite numbers, it is right (example $N \cdot 1/N$ with $N \rightarrow \infty$, is $0 \cdot \infty$, but from limits we know it is 1).

Thanks for reading! (x2)

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